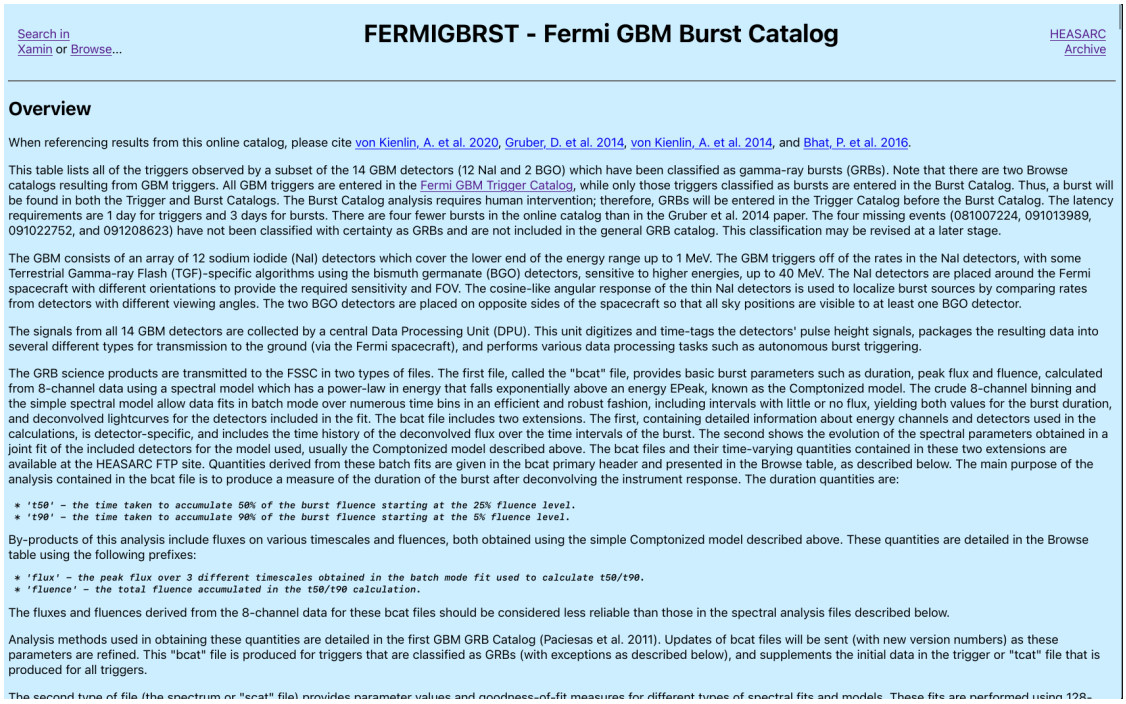
# Steps to download GBM data

1) Go to the URL : https://heasarc.gsfc.nasa.gov/W3Browse/fermi/fermigbrst.html



Search in Xamin or Browse...

**FERMIGBRST - Fermi GBM Burst Catalog**

HEASARC Archive

**Overview**

When referencing results from this online catalog, please cite von Kienlin, A. et al. 2020, Gruber, D. et al. 2014, von Kienlin, A. et al. 2014, and Bhat, P. et al. 2016.

This table lists all of the triggers observed by a subset of the 14 GBM detectors (12 NaI and 2 BGO) which have been classified as gamma-ray bursts (GRBs). Note that there are two Browse catalogs resulting from GBM triggers. All GBM triggers are entered in the Fermi GBM Trigger Catalog, while only those triggers classified as bursts are entered in the Burst Catalog. Thus, a burst will be found in both the Trigger and Burst Catalogs. The Burst Catalog analysis requires human intervention; therefore, GRBs will be entered in the Trigger Catalog before the Burst Catalog. The latency requirements are 1 day for triggers and 3 days for bursts. There are four fewer bursts in the online catalog than in the Gruber et al. 2014 paper. The four missing events (081007224, 091013989, 091022752, and 091208623) have not been classified with certainty as GRBs and are not included in the general GRB catalog. This classification may be revised at a later stage.

The GBM consists of an array of 12 sodium iodide (NaI) detectors which cover the lower end of the energy range up to 1 MeV. The GBM triggers off of the rates in the NaI detectors, with some Terrestrial Gamma-ray Flash (TGF)-specific algorithms using the bismuth germanate (BGO) detectors, sensitive to higher energies, up to 40 MeV. The NaI detectors are placed around the Fermi spacecraft with different orientations to provide the required sensitivity and FOV. The cosine-like angular response of the thin NaI detectors is used to localize burst sources by comparing rates from detectors with different viewing angles. The two BGO detectors are placed on opposite sides of the spacecraft so that all sky positions are visible to at least one BGO detector.

The signals from all 14 GBM detectors are collected by a central Data Processing Unit (DPU). This unit digitizes and time-tags the detectors' pulse height signals, packages the resulting data into several different types for transmission to the ground (via the Fermi spacecraft), and performs various data processing tasks such as autonomous burst triggering.

The GRB science products are transmitted to the FSSC in two types of files. The first file, called the "bcat" file, provides basic burst parameters such as duration, peak flux and fluence, calculated from 8-channel data using a spectral model which has a power-law in energy that falls exponentially above an energy EPeak, known as the Comptonized model. The crude 8-channel binning and the simple spectral model allow data fits in batch mode over numerous time bins in an efficient and robust fashion, including intervals with little or no flux, yielding both values for the burst duration, and deconvolved lightcurves for the detectors included in the fit. The bcat file includes two extensions. The first, containing detailed information about energy channels and detectors used in the calculations, is detector-specific, and includes the time history of the deconvolved flux over the time intervals of the burst. The second shows the evolution of the spectral parameters obtained in a joint fit of the included detectors for the model used, usually the Comptonized model described above. The bcat files and their time-varying quantities contained in these two extensions are available at the HEASARC FTP site. Quantities derived from these batch fits are given in the bcat primary header and presented in the Browse table, as described below. The main purpose of the analysis contained in the bcat file is to produce a measure of the duration of the burst after deconvolving the instrument response. The duration quantities are:

* *'t50' – the time taken to accumulate 50% of the burst fluence starting at the 25% fluence level.*
* *'t90' – the time taken to accumulate 90% of the burst fluence starting at the 5% fluence level.*

By-products of this analysis include fluxes on various timescales and fluences, both obtained using the simple Comptonized model described above. These quantities are detailed in the Browse table using the following prefixes:

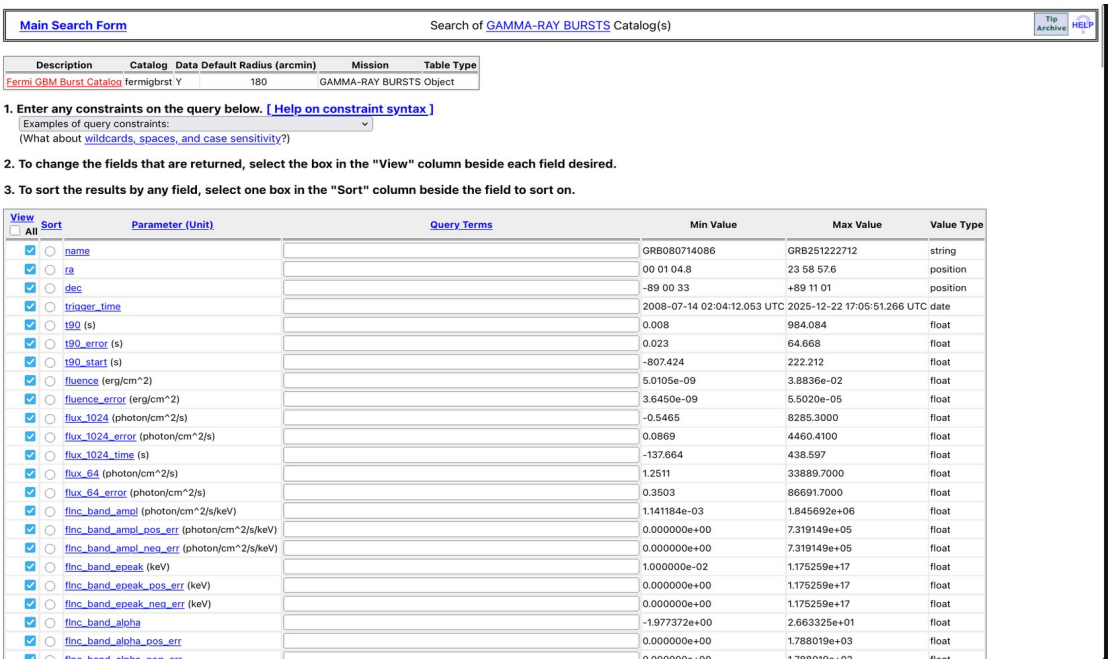
* *'flux' – the peak flux over 3 different timescales obtained in the batch mode fit used to calculate t50/t90.*
* *'fluence' – the total fluence accumulated in the t50/t90 calculation.*

The fluxes and fluences derived from the 8-channel data for these bcat files should be considered less reliable than those in the spectral analysis files described below.

Analysis methods used in obtaining these quantities are detailed in the first GBM GRB Catalog (Paciesas et al. 2011). Updates of bcat files will be sent (with new version numbers) as these parameters are refined. This "bcat" file is produced for triggers that are classified as GRBs (with exceptions as described below), and supplements the initial data in the trigger or "tcat" file that is produced for all triggers.

The second type of file (the spectrum or "scat" file) provides parameter values and goodness-of-fit measures for different types of spectral fits and models. These fits are performed using 128-

2) Click on Browse on the top-left corner of the page.
3) It will take you to following page URL : https://heasarc.gsfc.nasa.gov/db-perl/W3Browse/w3table.pl?tablehead=name%3Dfermigbrst&Action=More+Options



Main Search Form — Search of GAMMA-RAY BURSTS Catalog(s)

| Description | Catalog | Data | Default Radius (arcmin) | Mission | Table Type |
|---|---|---|---|---|---|
| Fermi GBM Burst Catalog | fermigbrst | Y | 180 | GAMMA-RAY BURSTS | Object |

**1. Enter any constraints on the query below.** [ Help on constraint syntax ]

Examples of query constraints:

(What about wildcards, spaces, and case sensitivity?)

**2. To change the fields that are returned, select the box in the "View" column beside each field desired.**

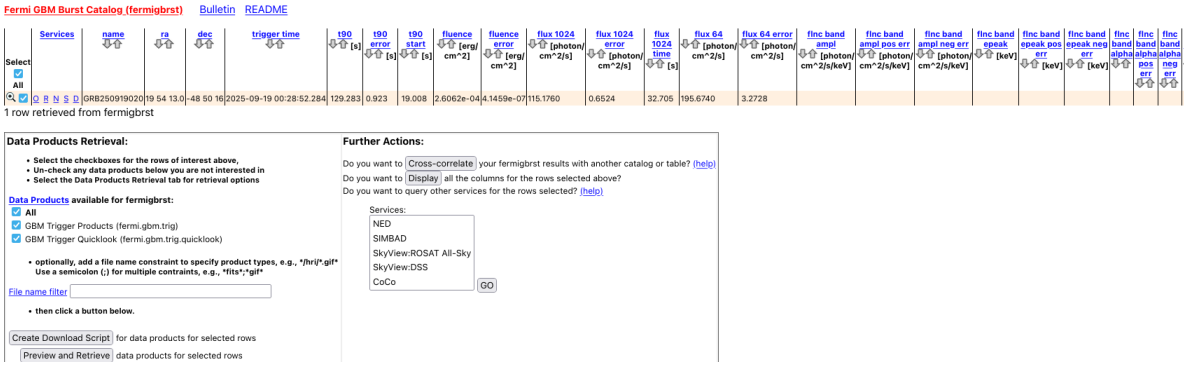
**3. To sort the results by any field, select one box in the "Sort" column beside the field to sort on.**

| View (All) | Sort | Parameter (Unit) | Query Terms | Min Value | Max Value | Value Type |
|---|---|---|---|---|---|---|
| ☑ | ○ | name | | GRB080714086 | GRB251222712 | string |
| ☑ | ○ | ra | | 00 01 04.8 | 23 58 57.6 | position |
| ☑ | ○ | dec | | -89 00 33 | +89 11 01 | position |
| ☑ | ○ | trigger_time | | 2008-07-14 02:04:12.053 UTC | 2025-12-22 17:05:51.266 UTC | date |
| ☑ | ○ | t90 (s) | | 0.008 | 984.084 | float |
| ☑ | ○ | t90_error (s) | | 0.023 | 64.668 | float |
| ☑ | ○ | t90_start (s) | | -807.424 | 222.212 | float |
| ☑ | ○ | fluence (erg/cm^2) | | 5.0105e-09 | 3.8836e-02 | float |
| ☑ | ○ | fluence_error (erg/cm^2) | | 3.6450e-09 | 5.5020e-05 | float |
| ☑ | ○ | flux_1024 (photon/cm^2/s) | | -0.5465 | 8285.3000 | float |
| ☑ | ○ | flux_1024_error (photon/cm^2/s) | | 0.0869 | 4460.4100 | float |
| ☑ | ○ | flux_1024_time (s) | | -137.664 | 438.597 | float |
| ☑ | ○ | flux_64 (photon/cm^2/s) | | 1.2511 | 33889.7000 | float |
| ☑ | ○ | flux_64_error (photon/cm^2/s) | | 0.3503 | 86691.7000 | float |
| ☑ | ○ | flnc_band_ampl (photon/cm^2/s/keV) | | 1.141184e-03 | 1.845692e+06 | float |
| ☑ | ○ | flnc_band_ampl_pos_err (photon/cm^2/s/keV) | | 0.000000e+00 | 7.319149e+05 | float |
| ☑ | ○ | flnc_band_ampl_neg_err (photon/cm^2/s/keV) | | 0.000000e+00 | 7.319149e+05 | float |
| ☑ | ○ | flnc_band_epeak (keV) | | 1.000000e-02 | 1.175259e+17 | float |
| ☑ | ○ | flnc_band_epeak_pos_err (keV) | | 0.000000e+00 | 1.175259e+17 | float |
| ☑ | ○ | flnc_band_epeak_neg_err (keV) | | 0.000000e+00 | 1.175259e+17 | float |
| ☑ | ○ | flnc_band_alpha | | -1.977372e+00 | 2.663325e+01 | float |
| ☑ | ○ | flnc_band_alpha_pos_err | | 0.000000e+00 | 1.788019e+03 | float |

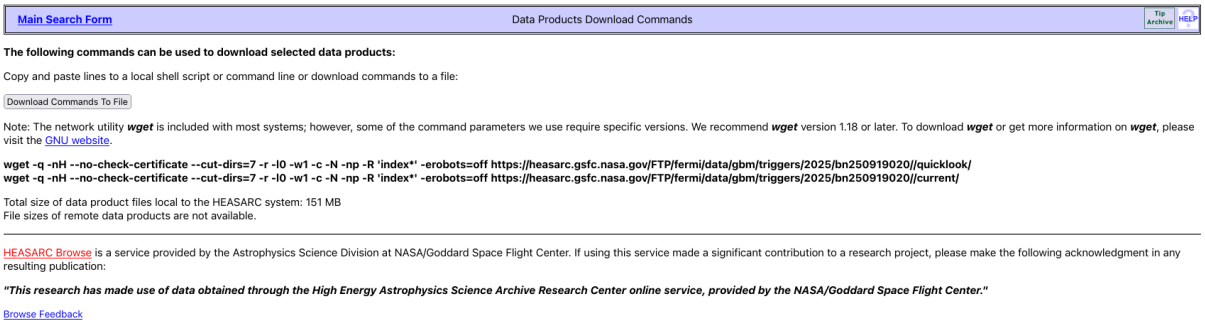
4) Enter the name of GRB , select all below view option and press Enter.
   Format : GRBYYMMDDfff e.g. GRB250919020

(fff: fraction of the day). It can be found by searching GRB name GCN Circular on Google and find line like `(trigger 779934537/250919020))`.

5) Click on Select all and then click on create download script.




6) Copy the wget link ( Current one).




7) Now make a GRB Folder in your PC and open Terminal there and copy paste the wget url and execute it. It will download all the files in that folder.